Week 7: Randomization

Empirical Research Methods

Nils Handler

University of Zurich (UZH)

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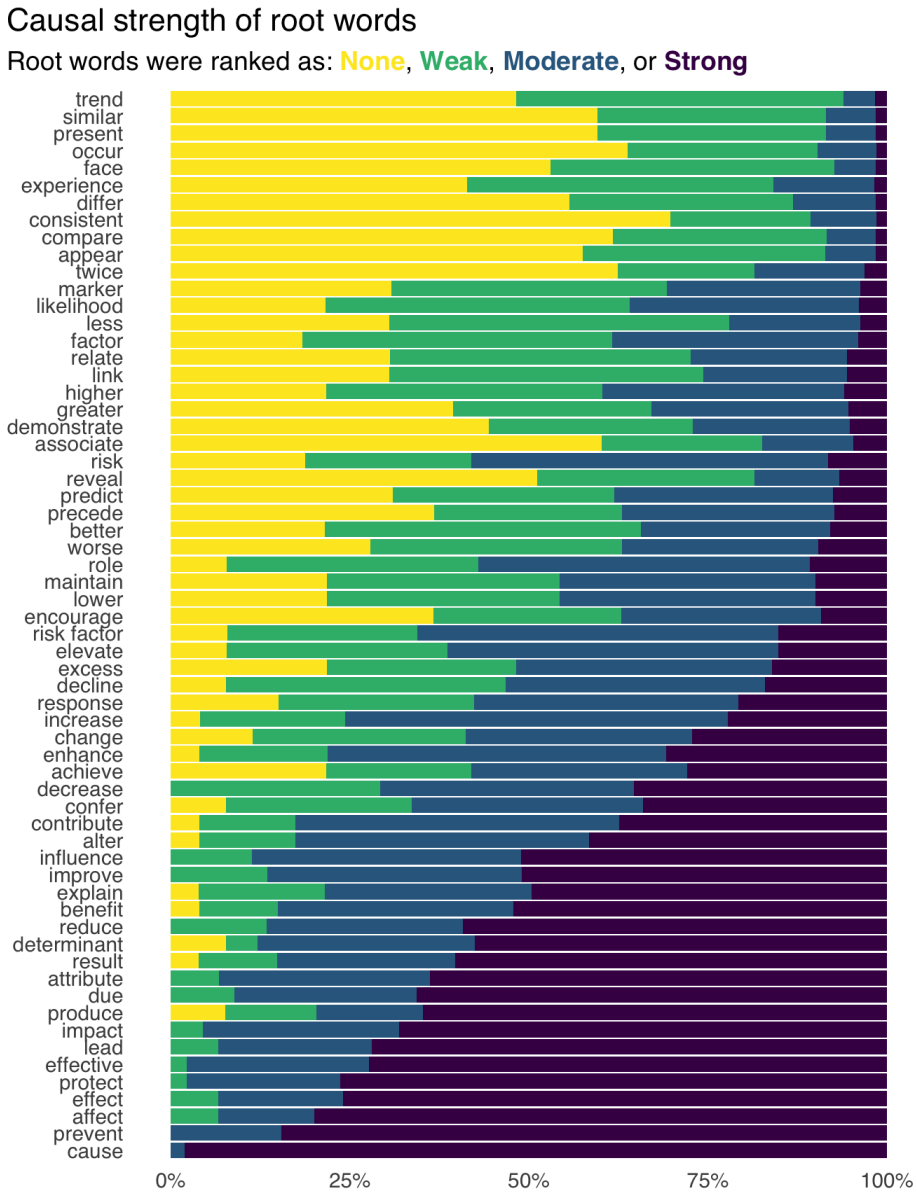
Learning Objectives

- Understand why randomization solves the confounding problem
- Learn the mathematical foundation of randomization
- Explore different types of randomization strategies
- Understand the advantages and disadvantages of each approach
- Connect randomization to your pre-analysis plan

The Importance of Causality

- Causality is sought after by researchers in their work, as it allows for stronger claims and the investigation of mechanisms.
- Economics currently has a strong preference towards causal methods in research.
- This has led to a discussion in the language surrounding causality as well.

- **Today**: Randomization helps ensure high-quality causal estimates



Motivation for Randomization

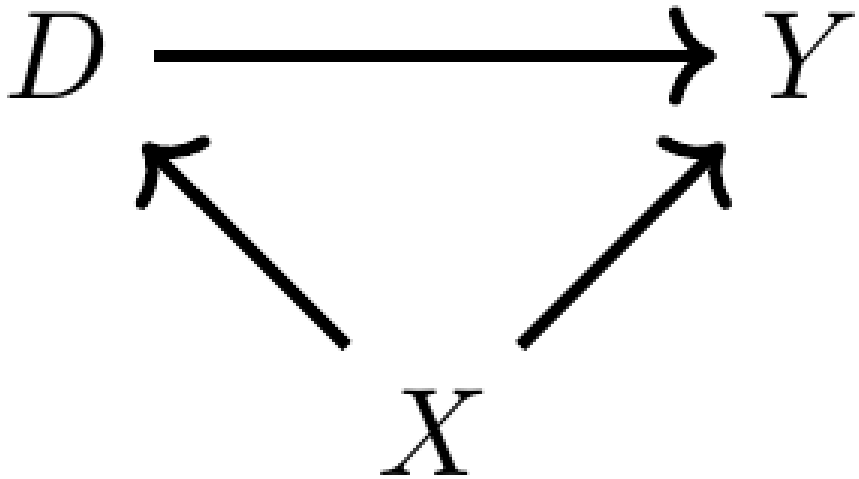
The Core Problem

Observational data suffers from confounding: unobserved variables affect both treatment and outcome.

How can we solve this problem?

Randomization: By randomly assigning treatment, we break the link between confounders and treatment assignment by setting the values of the treatment variables ourselves.

Randomization on a DAG



What does it mean for our DAG if we randomize on a variable (for example, D)?

It means that we break off all relationships to the parents, since we will fix the value of that node (typically the treatment).

We do need all the variables to be observed to actually know the *causal* distribution/model. Otherwise, we will still be missing the impact of the unobserved confounder. See Shalizi n.d. for more.

Randomization and Balancing

When we randomize our units into different groups, we risk creating groups with different characteristics. Ensuring groups have the same characteristics across covariates that may impact our outcome is called **balancing**.

We check for balancing once we have randomized the units into different groups by creating a balance table along our covariates of interest.

Can we drag a DAG for the model above?

	Intervention	Control
City 1, ages 5-7	23	24
City 2, ages 8-9	31	31
City 3, ages 10-11	28	27
City 1, ages 8-9	19	20
Cityl 2, ages 10-11	25	24
City 3, ages 5-7	35	35
City 1, ages 10-11	22	27
City 2, ages 5-7	24	23
City 3, ages 8-9	31	31

Bishop and Thompson 2023

Common Types of Randomization

1. Simple/Permutation Randomization

Each unit independently assigned with fixed probability. **Risk:** Imbalanced groups by chance.

2. Stratified Randomization

Stratify by important variables, randomize within strata. **Benefit:** Ensures balance on key characteristics.

3. Cluster Randomization

Randomize groups/clusters instead of individuals. **Benefit:** Reduces spillovers.

Simple/Permutation Randomization

Definition

Each unit has an equal probability of being assigned to treatment or control.

Mathematical Examples:

- **Simple**: Flip a fair coin for each participant, heads = treatment (T=1), probability of treatment: $P(T = 1) = 0.5$
- **Permutation**: Urn with 50 balls of two colours, one colour for each arm, draw balls without replacement, probability of treatment is still 0.5 overall

Pros:

- Easy to implement
- Balances the population "on average"

Cons:

- Risks imbalanced groups based on confounding characteristics
- Does not take advantage of known differences

Stratified Randomization

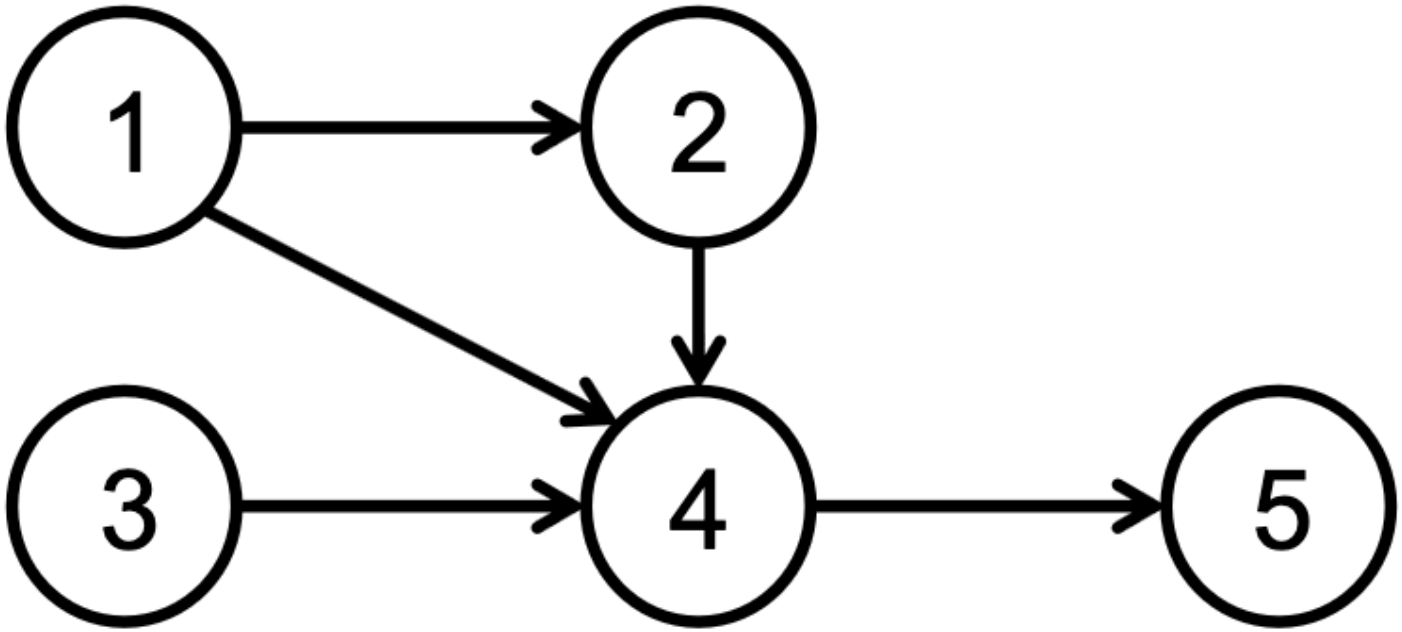
Definition

Along a set of known variables which can influence the outcome, participants grouped by these characteristics, then randomized *within* these groups.

Stratification allows us to examine how the treatment varies under other variables that the outcome and ensure the effects of these variables are balanced across different covariates.

Stratification can increase power by ensuring covariate impacts on the outcome are more evently distributed and thus making the impact of the treatment more clear/precise. If the differences are very small, however, this benefit will be reduced.

Stratification on a DAG



How might we want to stratify on this plot if our treatment variable is 2 and the outcome variable is 4? **Stratify based on 1 and 3.**

Stratification Example

Basic Example: Energy Efficiency Program

Suppose we have 350 units on which to randomize between an energy efficiency program and a control group. We know these units vary on income. How could we use this information to ensure we are learning as much as possible about the treatment?

Stratum	**Units**	**Treatment**	**Control**
Low Income	100	50	50
Middle Income	150	75	75
High Income	100	50	50
Total	**350**	**175**	**175**

Advantages of Stratified Randomization

1. Improved Precision

By balancing on predictive variables, we reduce variance in our estimates.

2. Subgroup Analysis

We can reliably estimate treatment effects for each stratum.

3. Ensures Balance

Guarantees that treatment and control are balanced on stratification variables.

4. Addresses Practical Constraints

Ensures sufficient sample size in each stratum for subgroup analysis.

Cluster Randomization

Definition

Units are taken in groups that are similar then assigned together to a specific treatment arm.

Why Cluster?

When treatment can spill over to nearby units, randomize at the cluster level (e.g., neighborhoods, schools).

Energy Saving Tips Example: For collecting data in a town, each neighborhood could act as a cluster. Each neighborhood is then assigned *entirely* to a specific treatment arm or control.

Design Effect in Cluster Randomization

The Design Effect

Clustering reduces effective sample size due to internal correlation of the units:

$$\text{Design Effect} = 1 + (m - 1)\rho$$

where m = cluster size, ρ = intra-cluster correlation.

Implication: You need a larger sample to achieve the same power, as the "effective" information we get from each new unit is lower from their correlation.

Example:

- 100 clusters, 10 people per cluster
- Intra-cluster correlation: $\rho = 0.05$
- Design Effect: $1 + (10 - 1) \times 0.05 = 1.45$
- Effective sample size: $1000/1.45 \approx 690$

Comparing Randomization Strategies

Feature	**Simple**	**Stratified**	**Clustered**
Complexity	Low	Medium	Medium
Balance on predictors	Not guaranteed	Guaranteed	Not guaranteed
Precision	Lower	Higher	Lower
Subgroup analysis	Risky	Reliable	Very Risky
Spillover control	None	None	Reliable
When to use	Large samples	Smaller samples	Logistical concerns

Practical Uses of Randomization

Biases	Remedies
Spontaneous improvement	Control group
Practice effects	Control group
Regression to the mean	Control group
Noisy data (1)	Strict selection criteria for participants
Noisy data (2)	Outcome measures with low measurement error
Selection bias	Random assignment to intervention
Placebo effects	Participant unaware of assignment
Experimenter bias (1)	Experimenter unaware of assignment
Experimenter bias (2)	Strictly specified protocol
Biased drop-outs	Intention-to-treat analysis
Low power	A priori power analysis
False positives due to p-hacking	Registration of trial protocol

Additional Methods in the Literature

- Propensity scores: Converting all observed covariates into a single score and pairing units together with similar scores, such that they are then split into the different arms **bauer_87_2020**.
- Block randomization: Blocking typically refers to splitting along naturally-occuring partitions in the data and randomizing within these splits, such as plots of land Meier 2026.
- Covariate-adaptive randomization: Sequentially assigning units directly to ensure balance Kang, Ragan, and Park 2026.

See Chapter 7 in Bishop and Thompson 2023 for more details.

Practical Implementation

1. **Define the randomization unit:** Can it match the observational unit?
2. **Choose the randomization type:** What biasing risks could we face?
3. **Identify stratification variables:** What predicts your outcome?
4. **Set the random seed:** For reproducibility.
5. **Generate randomization sequence:** Before seeing any data.
6. **Document everything:** Randomization procedure, seed, date, environment.
7. **Pre-register:** Publicly register your randomization plan.

Randomization in Your Pre-Analysis Plan

Your PAP Should Specify:

- **Randomization unit and type:** Who/what gets randomized?
- **Stratification/clustering variables:** If applicable, which variables and why?
- **Randomization ratio:** 1:1, 2:1, or other?
- **Reproducibility elements:** Seed, environment, etc.
- **Timing:** When will randomization occur?

Randomization Checks in Literature Reviews

You can use the quality of the randomization as a criteria for inclusion of a study. Using what we have covered, what are some important items to check for to ensure randomization was of high quality?

- Check for balance between covariates
- Examine if covariates that could impact the outcome are addressed
- Check DAG (if available) and see if all variables are dealt with or addressed through randomization or explanation

Good Practices in Randomization

- **Pre-register:** Document your randomization plan before implementation
- **Use a random seed:** Ensures reproducibility
- **Blind when possible:** Researchers shouldn't know assignments in advance
- **Check balance:** Verify that treatment and control are balanced
- **Document deviations:** If randomization doesn't go as planned, document why
- **Transparency:** Share your randomization code and procedure
- **Intention-to-Treat:** Analyze by original assignment, not actual treatment

Common Challenges in Randomization

Challenge 1: Non-Compliance

Some units don't receive assigned treatment. Solution: Use Intention-to-Treat analysis.

Challenge 2: Attrition

Units drop out of the study. Solution: Monitor attrition, use sensitivity analysis.

Challenge 3: Spillovers

Treatment affects control units. Solution: Use cluster randomization.

Challenge 4: Imbalance by Chance

Random assignment creates imbalance. Solution: Use stratified randomization.

Summary: Randomization

Key Concepts

- Randomization breaks the link between confounders and treatment
- Different randomization strategies serve different purposes
- Stratified randomization improves precision and ensures balance
- Cluster randomization reduces spillovers but requires larger samples

Next Steps

- Choose your randomization strategy based on your context
- Identify stratification variables if using stratified randomization
- Document your randomization plan in your pre-analysis plan
- Pre-register your randomization procedure

Citations and Resources I

- Bishop, Dorothy V. M. and Paul A. Thompson (Mar. 2023). *Evaluating What Works*. URL: https://bookdown.org/dorothy_bishop/Evaluating_What_Works/ (visited on 03/23/2026).
- Heinze-Deml, Christina (2026). *04_covariate_adjustment_I_0804.pdf*. en. URL: https://polybox.ethz.ch/index.php/s/eWwgUzMIuGJG5jz (visited on 04/13/2026).
- Kang, Minsoo, Brian G. Ragan, and Jae-Hyeon Park (2026). "Issues in Outcomes Research: An Overview of Randomization Techniques for Clinical Trials". en. In: *Journal of Athletic Training* 43.2 (), pp. 215–221. ISSN: 1062-6050. DOI: 10.4085/1062-6050-43.2.215. URL: https://nata.kglmeridian.com/view/journals/attr/43/2/article-p215.xml (visited on 04/13/2026).

Citations and Resources II

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- Shalizi, Cosma Rohilla (n.d.). "Advanced Data Analysis from an Elementary Point of View". en. In: ().